Midterm Study Guide

Introductory Psychology

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1 Introduction

This is the instructor-provided study guide for the midterm exam for PSY-101. Please use this document as **one** of the aids in preparing for this exam, but I strongly encourage you to develop additional study aids and tools on your own. This document is **not meant to be comprehensive** to all of the content covered in modules 1 through 7 - it is a merely a summary of what was covered, and meant as a review tool to help guide your practice.

Midterm / Exam 1 Structure covers the basic policies and procedures of the test, with information lifted directly from the syllabus. Tips for Preparing highlights what I feel are good practices to employ as you get ready to take the test. Tips for Studying contains ideas on what additional steps you can take to help get ready for the exam and [Responsible Use of AI Tools for Studying]. Vocabulary contains many (but not necessarily all) of the important terms and phrases used in the first 7 modules, and Learning Objectives has the all of the learning objectives for each module, as they appeared on each weekly lecture and overview.

Please do feel free to reach out with any questions you may have, but also make ready use of these many opportunities and resources to do well. I believe in each of you to use your time and diligence to turn in an excellent performance on this test.

2 Midterm / Exam 1 Structure

Please review the following information and procedures regarding the midterm and final exams in this course. The following text is also in the syllabus.

There may be additional section-specific policies relevant to the exam; please check the syllabus for the most accurate description of the exam as it is relevant to your section and reach out to the instructor via email with any additional questions

There will be 2 exams in this course, effectively a midterm and a final. These exams are intended to be **non-cumulative** and will cover content from all units and quizzes since the prior test. In practice, this means that the midterm will only cover the first half of the course, and the final will only cover the second half of the course.

The format is as follows:

- Each exam is 50 multiple-choice questions, 2 points for each question
- Exams will contain content from the entire unit, between all lectures AND readings and any other activities
- Exams are timed, 60 minutes only (used to be 50 minutes)
- Exams are not open-note, open-book, or collaborative. You are not permitted to use any form of assistance other than what is explicitly provided by the instructor. Do not discuss the tests with other students after it has concluded
- You should not be speaking, texting, or calling with other people during the exam. You should not be holding, using, or referring to any phone, tablet, or other electronic

device during the exam. You should not be wearing any headphones, earbuds, headsets, or other ear devices (unless it is pre-approved as an accommodation for medical reasons)
- Failure to adhere to the above guidelines may result in an automatic failure on the exam, if it raises the suspicion of un-authorized help during the exam, at the discretion of the instructor. Multiple failures to adhere to the rules of the exam may result in dismissal from class and referral to GVSU administration for further disciplinary action.
- Exams will be graded promptly and reviewed as soon as possible; the correct answers to the questions will also be provided.

3 Tips for Preparing

The below pieces of advice are my opinionated thoughts about how you may successfully prepare for this exam. Each student may find different strategies and techniques useful, please customize the following advice to your own needs.

- Because the exam is fairly comprehensive, you should make sure to review any materials relevant to the content we've covered in class, which may include:
  - Any and all lecture notes taken, whether that be the guided notes filled out or your own personal notes
  - Printed copies of your lecture check-ins
  - Any notes or activities you've completed from the book as additional practice
  - Any additional handwritten or student-typed preparation, such as answers to this study guide
- If you suspect that you may be absent or miss the exam window, you should contact the instructor as soon as possible
- If you have a documented accommodation for this course, you should work with the instructor to ensure that the exam format will be adequately address your needs, and work to make other arrangements for taking the test if necessary
- Try to get a good night of sleep for the several nights leading up to the exam
- Follow your usual diet leading up to the exam; try not to eat new foods that may upset your stomach and/or disrupt your attention while you attempt to exam
- If you normally consume it, keep your usual habits of consuming a healthy dose of caffeine leading up to the exam
- Dress in layers that you can add or take off to remain comfortable temperature-wise, during testing
  - However, do not wear any hoods up or hats that obscure your ears and may raise concern that you are wearing any earbuds

The below bullet points are only relevant to online sections of this course:

- You should take the exam on a laptop or desktop computer, not a tablet or phone. You should have your computed plugged in during the exam to reduce risk of battery running out, and should warn others you live with to not disturb you until you are done
- Find a quiet, secluded spot with good internet connection and an outlet for your computer. You may want to take the exam in a different place than your dorm room, if you feel that your roommates may not give you adequate private space to take the exam
- Place any other devices and distractions (e.g., phone, earbuds, etc.) in another room or away in a bag, unless necessary for medical or academic accommodations
- Please do not have any snacks or drinks while taking the exam, as their packaging may raise the concerns of cheating. Keep your workspace entirely clear of any materials or devices that could be perceived as cheating

4 Tips for Studying

The below pieces of advice are my opinionated thoughts about how you may successfully study for this exam. Each student may find different strategies and techniques useful, please customize the following advice to your own needs.

- As a general rule: **create, don't consume!** The best study materials are the ones that you make and develop. While I give you this document as a starting point, you should make the notes and study materials that are best suited to *you*. The process of developing study materials on your own will help cement the content for you.
- **Give yourself enough time to study!** Good studying starts early, and remains consistent. There is simply too much material to making cramming an effective strategy. Starting studying only the night before an exam will likely be insufficient for getting you a good grade.
- **Be honest with yourself about your ability on the content!** Too many students seem to "gaslight" themselves into believing they "know" the content, just by re-reading their notes (I've done it myself). It never hurts to test or quiz yourself and admit when you struggle on a certain area or topic.
- **DO NOT use this study guide as your sole resource when studying.** I provided this guide as a convenience, but I make no certain guarantee that it contains any and all information relevant to the exam. Any of the content covered in lectures or reading is fair game to be tested upon - and you should use resources from those things to prepare. Completion of this study guide does not necessarily guarantee success on the actual exam.
- If you do not know where to start:
  - First, focus on the Vocabulary and being able to organically come up with definitions and explanations for those terms. This is where many students may

  opt to use flashcards or spaced repetition software like Quizlet or Anki. Make sure that when you are using those flashcards, that you are *testing* yourself, not merely reading the flashcards - this will help you memorize the concepts
  - Second, work through the Learning Objectives - and test yourself upon those: try to write out responses to each of those objectives *without* looking at your notes
  - Third, go back through and memorize the diagrams of the ear, eye, brain, and neuron (find these diagrams under their respective areas in Blackboard) - practice labeling each part of these without any assistance
  - Between these three steps: if you can give good definitions to all of the vocabulary, can address each of the learning objectives, and can label all of those diagrams without your notes or study aids, you are almost certainly ready to do well on the exam.

- Try "teaching" the materials to an (un)willing volunteer or inanimate object
  - Trying to teach through and explain a concept will immediately make it apparent whether you understand it, or do not. This is how I double-check myself when writing lectures: if I can't explain something fluidly, then I haven't truly mastered explaining it well enough yet.
  - You could even try using my slides and see if you can talk "between the lines" and go beyond what is just written down - lecturing on a certain piece of content can actually really help you reinforce your understanding.

- Review past quizzes and lecture check-in answers; ensure you have corrected your understanding where you may have gotten a question wrong the first time through

- Consider "re-annotating" your previous notes:
  - Add question marks where you are confused on something
  - Add highlights to parts that stick out as especially critical to understanding
  - Re-work discussion and multiple choice questions strewn throughout the lecture guided notes and make sure you have good answers for each of them

- Complete practice and review questions at the end of each chapter of the textbook, your textbook has many useful activities for thinking through and practicing the concepts of the class; many students do not realize how useful these practices can be for preparing

- Consider coming to office hours the instructor to clarify difficult topics

5 Responsible Uses of AI Tools in Studying

I generally caution against use of AI in studying and the educational process in general, because many students (in my experience) use it in a way that deprives them of learning, rather than enhancing it. *But,* I do believe there is a way to use it to assist in the learning and memorization process. Let me give some examples of what to do and what not to do:

- As a general rule, **always double-check with your notes and textbook to ensure you get accurate answers - AI IS NOT INFALLIBLE**
- Avoid using plain ‘explanation’ prompts with no follow-up:
  - E.g., “Explain behaviorism to me”, “Tell me the lobes of the brain”
  - While such a question may *feel* like it helps, you likely won’t actually remember what it says unless you use that explanation to do something like write a definition in your own words
  - This is a good starting point for clarifying things, but won’t help in the broad scheme of things past that
- Use it to generate tools like mnemonics or tricks for assisting memorization:
  - E.g., “Make me an easy-to-remember acronym or mnemonic for remembering the lobes of the brain and their general functions”
  - If you actually practice memorization with these shortcuts, you can likely really benefit from these!
- Use it to make some mock or test questions based on your notes:
  - E.g., Upload your notes and ask, “Make me some example multiple-choice and short answer questions based on this document, and give me explanations for each one”
  - May have mixed results depending on how well the AI can “read” your notes
  - Make sure you actually test yourself and correct your understanding, not just using the answer key to make you “think” you know the correct answer, when you can’t really explain it
- Use it to make short passage examples of certain principles in practice:
  - E.g., “Write me 4 examples of classical conditioning of behavior using positive reinforcement, negative reinforcement, positive punishment, and negative punishment”

6 Vocabulary

The terms in this section have been lifted from the textbook bolded terms and may not have been used directly in lecture, though I have tried to remove those not explicitly covered by my recordings. If you cannot find a clear definition or example, consider looking at the end of each chapter for the author-provided definition, or review the video from the respective module to see my explanation again.

Some terms may be duplicated due to having appeared as bolded terms in multiple chapters, consider their relevance to the context of each module.

6.1 Module 1

American Psychological Association (APA)
behaviorism
biopsychology
biopsychosocial model
clinical psychology
cognitive psychology
cognitivism
counseling psychology
developmental psychology
dissertation
empirical method
forensic psychology
functionalism
humanism
introspection
-ology
personality psychology
personality trait
PhD
postdoctoral training program
psychoanalytic theory / psychoanalysis
psychology
PsyD
sport and exercise psychology
structuralism

6.2 Module 2

archival research
attrition
cause-and-effect relationship
clinical or case study
confirmation bias
confounding variable
control group
correlation
correlation coefficient
cross-sectional research
debriefing
deception
deductive reasoning
dependent variable
double-blind study
empirical
experimental group
experimenter bias
fact
falsifiable
generalize
hypothesis
illusory correlation
independent variable
inductive reasoning
informed consent
Institutional Animal Care and Use Committee (IACUC)

Institutional Review Board (IRB)
inter-rater reliability
longitudinal research
naturalistic observation
negative correlation
observer bias
operational definition
opinion
participants
peer-reviewed journal article
placebo effect
population
positive correlation
random assignment
random sample
reliability
replicate
sample
single-blind study
statistical analysis
survey
theory
validity

6.3 Module 3

action potential
adrenal gland
agonist
all-or-none
allele
amygdala
antagonist
auditory cortex
autonomic nervous system
axon
biological perspective
Broca's area
central nervous system (CNS)
cerebellum
cerebral cortex
chromosome
computerized tomography (CT) scan
corpus callosum
dendrite
deoxyribonucleic acid (DNA)
diabetes
dominant allele
electroencephalography (EEG)
endocrine system
epigenetics
fight or flight response
forebrain
fraternal twins
frontal lobe

functional magnetic resonance imaging (fMRI)

gene

genetic environmental correlation

genotype

glial cell

gonad

gyrus

hemisphere

heterozygous

hindbrain

hippocampus

homeostasis

homozygous

hormone

hypothalamus

identical twins

lateralization

limbic system

longitudinal fissure

magnetic resonance imaging (MRI)

medulla

membrane potential

midbrain

motor cortex

mutation

myelin sheath

nervous system

neuron

neuroplasticity

neurotransmitter

Nodes of Ranvier

occipital lobe

pancreas

parasympathetic nervous system

parietal lobe

peripheral nervous system (PNS)

phenotype

pituitary gland

polygenic

pons

positron emission tomography (PET) scan

prefrontal cortex

psychotropic medication

range of reaction

receptor

recessive allele

resting potential

reticular formation

reuptake

semipermeable membrane

soma

somatic nervous system

somatosensory cortex

substantia nigra

sulcus

sympathetic nervous system

synaptic cleft

synaptic vesicle

temporal lobe

terminal button
thalamus
theory of evolution by natural selection
threshold of excitation
thyroid
ventral tegmental area (VTA)
Wernicke's area

6.4 Module 4

alpha wave
beta wave
biological rhythm
central sleep apnea
circadian rhythm
cognitive-behavioral therapy
consciousness
delta wave
evolutionary psychology
homeostasis
insomnia
jet lag
K-complex
latent content
lucid dream
manifest content
melatonin
meta-analysis
night terror
non-REM (NREM)
parasomnia
pineal gland
rapid eye movement (REM) sleep
REM sleep behavior disorder (RBD)
restless leg syndrome
rotating shift work
sleep
sleep apnea
sleep debt
sleep regulation
sleep spindle
sleepwalking
stage 1 sleep
stage 2 sleep
stage 3 sleep
sudden infant death syndrome (SIDS)
suprachiasmatic nucleus (SCN)
theta wave
wakefulness

6.5 Module 5

absolute threshold

amplitude

basilar membrane

binaural cue

binocular cue

binocular disparity

blind spot

bottom-up processing

closure

cochlea

cochlear implant

cone

congenital insensitivity to pain (congenital analgesia)

cornea

deafness

decibel (dB)

depth perception

electromagnetic spectrum

figure-ground relationship

fovea

frequency

hair cell

hertz (Hz)

inattentional blindness

incus

inflammatory pain

interaural level difference

interaural timing difference

iris

just noticeable difference

kinesthesia

lens

malleus

Meissner's corpuscle

Merkel's disk

monaural cue

monocular cue

neuropathic pain

nociception

olfactory bulb

olfactory receptor

opponent-process theory of color perception

optic chiasm

optic nerve

ossicles

Pacinian corpuscle

peak

perception

perceptual hypothesis

photoreceptor

pinna

pitch

place theory of pitch perception

proprioception

proximity

pupil

retina

rod

Ruffini corpuscle

sensation
sensorineural hearing loss
sensory adaptation
similarity
stapes
subliminal message
taste bud
temporal theory of pitch perception
thermoception
top-down processing
transduction
trichromatic theory of color perception
trough
tympanic membrane
umami
vertigo
vestibular sense
visible spectrum
wavelength

6.6 Module 6

acquisition
associative learning
classical conditioning
cognitive map
conditioned response (CR)
conditioned stimulus (CS)
continuous reinforcement
extinction
instinct
latent learning
law of effect
learning
model
negative punishment
negative reinforcement
neutral stimulus (NS)
observational learning
operant conditioning
positive punishment
positive reinforcement
primary reinforcer
punishment
radical behaviorism
reflex
reinforcement
secondary reinforcer
shaping
spontaneous recovery
stimulus discrimination
stimulus generalization
unconditioned response (UCR)
unconditioned stimulus (UCS)
vicarious punishment
vicarious reinforcement

6.7 Module 7

algorithm
analytical intelligence
anchoring bias
artificial concept
availability heuristic
cognition
cognitive psychology
cognitive script
concept
confirmation bias
convergent thinking
creative intelligence
creativity
crystallized intelligence
cultural intelligence
divergent thinking
dyscalculia
dysgraphia
dyslexia
emotional intelligence
event schema
fluid intelligence
Flynn effect
functional fixedness
grammar
heuristic
hindsight bias
intelligence quotient
language
lexicon
mental set
morpheme
Multiple Intelligences Theory
natural concept
norming
overgeneralization
phoneme
practical intelligence
problem-solving strategy
prototype
range of reaction
representative bias
representative sample
role schema
schema
semantics
standard deviation
standardization
syntax
trial and error
triarchic theory of intelligence
working backwards

7 Learning Objectives

This is a list of the learning objectives, as they appeared through the module overviews/check-list and lectures

7.1 Module 1

- Understand the importance of Wundt and James in the development of psychology
- Appreciate Freud's influence on psychology
- Understand the basic tenets of Gestalt psychology
- Appreciate the important role that behaviorism played in psychology's history
- Understand basic tenets of humanism
- Understand how the cognitive revolution shifted psychology's focus back to the mind
- Define psychology
- Understand the merits of an education in psychology
- Appreciate the diversity of interests and foci within psychology
- Understand basic interests and applications in each of the described areas of psychology
- Demonstrate familiarity with some of the major concepts or important figures in each of the described areas of psychology
- Understand educational requirements for careers in academic settings
- Understand the demands of a career in an academic setting
- Understand career options outside of academic settings
- Appreciate the application of psychology to wide array of scientific questions and areas
- Be able to reasonably discriminate between different perspectives and domains of psychology and explain why a certain study or question fits underneath a particular area
- Critically consider whether you align with a certain perspective or domain of psychology

7.2 Module 2

- Explain how scientific research addresses questions about behavior
- Discuss how scientific research guides public policy
- Appreciate how scientific research can be important in making personal decisions
- Describe the different research methods used by psychologists
- Discuss the strengths and weaknesses of case studies, naturalistic observation, surveys, and archival research
- Compare longitudinal and cross-sectional approaches to research
- Compare and contrast correlation and causation
- Explain what a correlation coefficient tells us about the relationship between variables

- Recognize that correlation does not indicate a cause-and-effect relationship between variables
- Discuss our tendency to look for relationships between variables that do not really exist
- Explain random sampling and assignment of participants into experimental and control groups
- Discuss how experimenter or participant bias could affect the results of an experiment
- Identify independent and dependent variables
- Discuss how research involving human subjects is regulated
- Summarize the processes of informed consent and debriefing
- Explain how research involving animal subjects is regulated
- Understand the critical role research plays in solidifying psychology as a science
- Understand the pitfalls and dangers of unethical research
- Be able to identify the core components and features of a described research design

7.3 Module 3

- Explain the basic principles of the theory of evolution by natural selection
- Describe the differences between genotype and phenotype
- Discuss how gene-environment interactions are critical for expression of physical and psychological characteristics
- Identify the basic parts of a neuron
- Describe how neurons communicate with each other
- Explain how drugs act as agonists or antagonists for a given neurotransmitter system
- Describe the difference between the central and peripheral nervous systems
- Explain the difference between the somatic and autonomic nervous systems
- Differentiate between the sympathetic and parasympathetic divisions of the autonomic nervous system
- Explain the functions of the spinal cord
- Identify the hemispheres and lobes of the brain
- Describe the types of techniques available to clinicians and researchers to image or scan the brain
- Identify the major glands of the endocrine system
- Identify the hormones secreted by each gland
- Describe each hormone's role in regulating bodily functions
- Understand the purpose on learning anatomy to contextualize human behavior and cognition
- Appreciate the complexity in our nervous system processes and how the interact to rise to cognition

7.4 Module 4

- Understand what is meant by consciousness

- Explain how circadian rhythms are involved in regulating the sleep-wake cycle, and how circadian cycles can be disrupted
- Discuss the concept of sleep debt
- Describe areas of the brain involved in sleep
- Understand hormone secretions associated with sleep
- Describe several theories aimed at explaining the function of sleep
- Name and describe three theories about why we dream
- Differentiate between REM and non-REM sleep
- Describe the differences between the three stages of non-REM sleep
- Understand the role that REM and non-REM sleep play in learning and memor
- Describe the symptoms and treatments of insomnia
- Recognize the symptoms of several parasomnias
- Describe the symptoms and treatments for sleep apnea
- Recognize risk factors associated with sudden infant death syndrome (SIDS) and steps to prevent it
- Describe the symptoms and treatments for narcolepsy
- Describe the diagnostic criteria for substance use disorders
- Identify the neurotransmitter systems impacted by various categories of drugs
- Describe how different categories of drugs affect behavior and experience
- Define hypnosis and meditation
- Understand the similarities and differences of hypnosis and meditation
- Understand how consciousness is better described as range of states, rather than simply “on” or “off”
- Know several common conditions and drugs that modify or otherwise change the state of consciousness

7.5 Module 5

- Distinguish between sensation and perception
- Describe the concepts of absolute threshold and difference threshold
- Discuss the roles attention, motivation, and sensory adaptation play in perception
- Describe important physical features of wave forms
- Show how physical properties of light waves are associated with perceptual experience
- Show how physical properties of sound waves are associated with perceptual experience
- Describe the basic anatomy of the visual system
- Discuss how rods and cones contribute to different aspects of vision
- Describe how monocular and binocular cues are used in the perception of depth
- Describe the basic anatomy and function of the auditory system
- Explain how we encode and perceive pitch
- Discuss how we localize sound
- Describe the basic functions of the chemical senses
- Explain the basic functions of the somatosensory, nociceptive, and thermoceptive

sensory systems
- Describe the basic functions of the vestibular, proprioceptive, and kinesthetic sensory systems
- Explain the figure-ground relationship
- Define Gestalt principles of grouping
- Describe how perceptual set is influenced by an individual's characteristics and mental state
- Be able to connect sensation and perception back to concepts in biopsychology, consciousness/sleep, and gestalt psychology
- Explain how the anatomical structure gives rise to functions of perception, much like we did on focusing with brain structures and their respective functions

7.6 Module 6

- Explain how learned behaviors are different from instincts and reflexes
- Define learning
- Recognize and define three basic forms of learning—classical conditioning, operant conditioning, and observational learning
- Explain how classical conditioning occurs
- Summarize the processes of acquisition, extinction, spontaneous recovery, generalization, and discrimination
- Define operant conditioning
- Explain the difference between reinforcement and punishment
- Distinguish between reinforcement schedules
- Define observational learning
- Discuss the steps in the modeling process
- Explain the prosocial and antisocial effects of observational learning
- Be able to discriminate between the 3 forms of learning given a detailed example
- Be able to connect learning back to the historical paradigm of behaviorism
- Be able to describe the process and nuances of each of the 3 types of learning

7.7 Module 7

- Describe cognition
- Distinguish concepts and prototypes
- Explain the difference between natural and artificial concepts
- Describe how schemata are organized and constructed
- Define language and demonstrate familiarity with the components of language
- Understand the development of language
- Explain the relationship between language and thinking
- Describe problem solving strategies
- Define algorithm and heuristic
- Explain some common roadblocks to effective problem solving and decision making

- Define intelligence
- Explain the triarchic theory of intelligence
- Identify the difference between intelligence theories
- Explain emotional intelligence
- Define creativity
- Explain how intelligence tests are developed
- Describe the history of the use of IQ tests
- Describe the purposes and benefits of intelligence testing
- Describe how genetics and environment affect intelligence
- Explain the relationship between IQ scores and socioeconomic status
- Describe the difference between a learning disability and a developmental disorder
- Understand the key vocabulary used in intelligence, language, and cognition theories
- Appreciate the complexity in defining and measuring intelligence
- Understand some of the basic developmental processes in language and intelligence